

# Computational Complexity of Recursive Bisection for Redistricting

B.6 — Algorithm Track

Gio Della-Libera

*Apportionment Research Group*

May 2026

## Abstract

We analyse the computational complexity of recursive bisection for congressional redistricting on planar census-tract graphs. We prove that the balanced  $k$ -cut redistricting problem is NP-hard even on planar graphs with unit vertex weights ( $k = 2$ ), by reduction from Connected Planar Graph Bisection [Dyer and Frieze, 1985]. The hardness result concerns optimisation (finding the minimum-cut partition); feasibility (finding any valid partition) is trivially solvable for connected graphs. No polynomial-time algorithm is known to achieve better than  $O(\log n)$  approximation for balanced  $k$ -partition in general; METIS’s heuristic quality is empirically characterised across 50 states. The algorithm runs in  $O(\texttt{niter} \cdot n \log k)$  time with  $O(n \log k)$  space ( $\texttt{niter} = 100$  in experiments). Empirically, on  $k > 1$  states across three census years (2000, 2010, 2020), runtime scales as  $O(n^{1.07 \pm 0.03})$  — nearly linear. Single-district states ( $k = 1$ ) are excluded from the scaling analysis as they require no bisection. For California ( $n = 8,057$  tracts,  $k = 52$  districts), peak memory is under 3 MB at tract resolution. These results provide the complexity-theoretic foundation for the Districting Integrity Act’s use of METIS as the statutory partitioner: exact optimisation is intractable, making heuristic approximation not merely acceptable but necessary.

## 1 Introduction

Congressional redistricting requires partitioning a weighted planar graph into  $k$  connected, population-balanced parts while minimising total edge cut (shared boundary perimeter). The algorithmic literature has long recognised that this problem is NP-hard in general [Garey and Johnson, 1979, Garey et al., 1976], motivating the use of heuristic partitioners such as METIS [Karypis and Kumar, 1998a]. However, the precise complexity status of the *redistricting variant* — with its planarity constraint, population-balance requirement, and boundary-length edge weights — has not been rigorously established.

Moreover, the empirical scaling behaviour of METIS recursive bisection on real census-tract graphs across three decades of census data has not been characterised. For the Districting

Integrity Act (DIA) to be legally defensible, practitioners and courts need to know: (1) is the optimisation problem NP-hard even for the planar case? (2) what approximation guarantee does METIS offer? (3) does the algorithm scale to large states and future Census enumerations?

## Contributions

This paper makes four contributions:

1. We establish NP-hardness of the balanced  $k$ -cut redistricting problem on planar graphs by reduction from Connected Planar Graph Bisection [Dyer and Frieze, 1985], correctly accounting for the contiguity (connectivity) requirement of the redistricting problem.
2. We clarify the approximation-theoretic landscape: no formal approximation ratio is known for METIS’s heuristic; the best polynomial-time bound ( $O(\sqrt{\log n})$  via ARV [Arora et al., 2009]) applies to Sparsest Cut, not to balanced connected  $k$ -partition. We characterise METIS’s empirical quality instead.
3. We provide a runtime analysis:  $O(\texttt{niter} \cdot n \log k)$  time with  $O(n \log k)$  space ( $\texttt{niter} = 100$  in experiments).
4. We fit the empirical scaling law  $T = a \cdot n^b$  to  $k > 1$  states across three census years, finding  $b = 1.07 \pm 0.03$  (near-linear); single-district states are excluded from the fit.

## Why complexity matters for redistricting law

The DIA’s algorithm specification needs a complexity-theoretic foundation for two reasons. First, if the redistricting problem were polynomial-time solvable, the DIA would be obligated to specify an exact algorithm rather than a heuristic. Our NP-hardness result shows that no such exact algorithm can exist unless  $P = NP$  (a hypothesis universally believed to be false, making the result effectively unconditional in practice). Second, while no formal approximation guarantee is established for METIS’s heuristic, the empirical characterisation in Section 4 shows that METIS consistently finds solutions within 3% of the best available partition [Della-Libera, 2026], providing strong practical evidence for legal challenges alleging systematic suboptimality. Importantly, the hardness result concerns the *optimisation* problem; a valid redistricting plan always exists and can be found efficiently — only finding the *optimal* one is intractable.

## Paper structure

Section 3 formally defines the redistricting problem, establishes NP-hardness, and characterises the approximation landscape. Section 4 analyses runtime complexity, empirical scaling, and space requirements. Section 5 discusses implications for the DIA. Section 6 concludes.

## 2 Related Work

### 2.1 Complexity of graph partitioning

The minimum bisection problem — partition a graph into two equal-weight parts minimising the edge cut — is NP-hard for general graphs [Garey et al., 1976]. Garey, Johnson, and Stockmeyer [Garey et al., 1976] showed that minimum bisection remains NP-hard even for degree-3 planar graphs via reduction from 3-SAT. Their reduction, however, does not impose connectivity on the two parts, and hence does not directly cover the redistricting variant, which requires each district to induce a connected subgraph.

Dyer and Frieze [Dyer and Frieze, 1985] proved that the problem of partitioning a graph into  $k$  *connected* subgraphs of prescribed sizes is NP-complete in general, and remains NP-complete for planar graphs. This is the result we use to anchor Theorem 1: the redistricting problem requires connected parts by definition, so the connected-partition variant is the correct source for the reduction.

The balanced  $k$ -partition problem (generalising bisection to  $k$  parts, without the connectivity requirement) is also NP-hard [Andreev and Räcke, 2006], and the minimum edge-cost  $k$ -cut problem is NP-hard for  $k \geq 3$  even without balance constraints [Goldschmidt and Hochbaum, 1994]. The connectivity requirement of the redistricting problem is at least as hard as the unconstrained case, since any algorithm for the unconstrained problem that happened to produce connected parts would solve the constrained problem.

### 2.2 Approximation algorithms

For general graphs, the best known approximation ratio for minimum bisection is  $O(\log n)$  [Andreev and Räcke, 2006]. For planar graphs, the Arora-Rao-Vazirani (ARV) framework [Arora et al., 2009] achieves  $O(\sqrt{\log n})$  approximation for Sparsest Cut via a semidefinite programming (SDP) relaxation. This bound applies to Sparsest Cut, not directly to Balanced  $k$ -Cut with contiguity; extending it to balanced connected  $k$ -partition with formal guarantees remains an open problem.

METIS [Karypis and Kumar, 1998a] achieves near-optimal results empirically on benchmark graphs but provides no formal approximation guarantee for balanced connected partition beyond the planar separator bound [Lipton and Tarjan, 1979]. The best known polynomial-time algorithm achieves no better than  $O(\log n)$  approximation for balanced  $k$ -partition in general; no known algorithm achieves a tighter formal bound for the connected variant. KaHyPar [Schlag et al., 2017] provides stronger guarantees on hypergraph instances but is not designed for planar adjacency graphs.

### 2.3 Complexity of redistricting

Discrete geometry approaches to redistricting [Duchin and Tenner, 2020] have characterised the solution space via Markov chains rather than complexity theory. Fifield et al. [Fifield et al., 2020] proved mixing bounds for redistricting Markov chains, showing that the solution space is well-connected — a result consistent with our seed sensitivity analysis (B.7).

Computational redistricting has been characterised as an optimisation problem by Mehrotra et al. [Mehrotra et al., 1998]. Neither work provides complexity results for the minimum edge-cut variant.

## 2.4 Runtime analysis of METIS

Karypis and Kumar [Karypis and Kumar, 1998a] proved  $O(m \log n)$  runtime per level for their multilevel scheme, where  $m = |E|$  and  $n = |V|$ . For planar graphs,  $m = O(n)$  (by Euler’s formula for planar graphs), giving  $O(n \log n)$  per bisection and  $O(n \log k \log n)$  overall for  $\lceil \log_2 k \rceil$  levels of bisection. Empirical scaling on non-planar graphs has been characterised for scientific computing workloads [Karypis and Kumar, 1998b], but not for census tract graphs with population-balance constraints.

## 3 Problem Formulation and Hardness

### 3.1 The redistricting graph model

Let  $G = (V, E, w_V, w_E)$  be a weighted planar graph where:

- $|V| = n$  is the number of census tracts,
- $w_V : V \rightarrow \mathbb{Z}_{>0}$  assigns population to each tract,
- $w_E : E \rightarrow \mathbb{R}_{>0}$  assigns shared boundary length to each edge, and
- $G$  is planar (tracts tile a surface without crossings).

**Problem 1** (Balanced  $k$ -Cut Redistricting). Given  $G = (V, E, w_V, w_E)$ , integer  $k \geq 2$ , and tolerance  $\varepsilon > 0$ , find a partition  $\mathcal{P} = \{P_1, \dots, P_k\}$  of  $V$  that:

1. minimises  $\text{EC}(\mathcal{P}) = \sum_{(u,v) \in E, \pi(u) \neq \pi(v)} w_E(u, v)$ ,
2. satisfies  $|w_V(P_i) - W/k| \leq \varepsilon W/k$  for all  $i$  (population balance), where  $W = \sum_v w_V(v)$ , and
3. each  $P_i$  induces a connected subgraph of  $G$  (contiguity).

### 3.2 NP-hardness of the redistricting problem

**Theorem 1** (NP-hardness). *Problem 1 is NP-hard for planar graphs, even with  $k = 2$  and unit vertex weights.*

*Proof sketch.* We reduce from the Connected Planar Graph Bisection problem, which asks whether a planar graph can be bisected into two connected parts of equal vertex count with edge cut at most  $C$ . This problem is NP-complete by the results of Dyer and Frieze [Dyer and Frieze, 1985] on connected graph partition on planar graphs; see also Garey and Johnson [Garey and Johnson, 1979] (Problem ND14) for the disconnected variant, whose connected restriction only strengthens hardness.

Given a Connected Planar Graph Bisection instance  $(G', C)$  with  $|V'|$  even, set all vertex weights  $w_V \equiv 1$  (unit population), all edge weights  $w_E \equiv 1$  (unit boundary),  $k = 2$ , and  $\varepsilon = 0$ . A valid solution to Problem 1 on this instance consists of two connected parts  $P_1, P_2$  with  $|P_1| = |P_2| = |V'|/2$  and total edge cut minimised — which is precisely a minimum connected bisection of  $G'$ . Conversely, every minimum connected bisection of  $G'$  yields a valid solution to Problem 1 (connectivity of parts satisfies contiguity; equal sizes satisfy population balance with  $\varepsilon = 0$ ). The reduction is a polynomial-time many-one reduction since all assignments are constant-time. The planar graph structure of census tract adjacency graphs is preserved under this reduction, as  $G'$  is planar by assumption and weight assignment does not alter planarity.  $\square$   $\square$

**Remark (feasibility vs. optimisation).** Theorem 1 establishes hardness of the *optimisation* variant (finding a partition with minimum edge cut). The *feasibility* variant — does any valid partition exist? — is trivially solvable for connected graphs with  $k \leq n$ : any spanning tree of  $G$  can be partitioned into  $k$  connected subtrees with appropriate edge removal. The NP-hardness result therefore concerns the difficulty of finding an *optimal* compact partition, not the existence of one.

**Proposition 1** (Approximation ratio). *No known polynomial-time algorithm achieves better than  $O(\log n)$  approximation for balanced  $k$ -partition in general. The approximation quality of METIS recursive bisection on redistricting-scale planar graphs is empirically characterised in Section 4; no formal approximation guarantee for METIS’s heuristic is claimed here.*

**Context.** The ARV framework [Arora et al., 2009] achieves  $O(\sqrt{\log n})$  approximation for Sparsest Cut via an SDP relaxation; this result applies to Sparsest Cut, not to Balanced  $k$ -Cut with contiguity. METIS’s Fiduccia–Mattheyses (FM) local search [Fiduccia and Mattheyses, 1982] has no formal connection to the ARV SDP bound. By the Lipton–Tarjan separator theorem [Lipton and Tarjan, 1979], every planar graph has a balanced separator of size  $O(\sqrt{n})$ , which gives an  $O(\sqrt{n})$  edge-cut upper bound for a single bisection; the approximation ratio this implies depends on the unknown optimum. Section 4 provides an empirical characterisation of how closely METIS tracks the best partition found across multiple seeds.

### 3.3 Empirical methodology

We measure runtime by executing the METIS recursive bisection (C FFI, version 5.1.0) on all 50 states across three census years (2000, 2010, 2020), with configurations:

- `ufactor = 5, niter = 100, seed = 42`
- Single-threaded execution on a 16-core AMD Ryzen 9 5950X
- Runtime measured via `std::time::Instant` in the Rust harness (`redist-cli`)
- Averaged over 10 independent runs to reduce variance

We fit  $T = a \cdot n^b$  using ordinary least squares on log-transformed data, reporting  $b \pm$  one standard error across the 50-state sample.

## 4 Runtime Analysis

### 4.1 Theoretical analysis

Each bisection step of METIS recursive bisection on a planar graph with  $|E_i|$  edges and  $|V_i|$  vertices involves:

1. Building the CSR graph representation:  $O(|E_i|)$  time
2. Heavy-edge matching for coarsening:  $O(|E_i|)$  time per level,  $O(\log |V_i|)$  coarsening levels
3. Greedy initial partitioning on the coarsened graph:  $O(|V_i|/c)$  for coarsening ratio  $c \approx 20$
4. Fiduccia–Mattheyses refinement [Fiduccia and Mattheyses, 1982]:  $O(|E_i| \cdot \mathbf{niter})$  per level

For a planar graph,  $|E_i| = O(|V_i|)$  at every level. The bisection tree has  $\lceil \log_2 k \rceil$  levels. At level  $\ell$ , there are  $2^\ell$  subproblems each of expected size  $O(n/2^\ell)$ . Total work at level  $\ell$ :

$$2^\ell \cdot O\left(\frac{n}{2^\ell} \cdot \log\left(\frac{n}{2^\ell}\right)\right) = O(n \log n).$$

Summing over  $\lceil \log_2 k \rceil$  levels:

$$T_{\text{RB}}(n, k) = O(n \log n \cdot \log k).$$

In the experiments,  $\mathbf{niter} = 100$  (a fixed constant). For  $n \leq 8,057$  (California),  $\log n \approx 13$ , so the FM refinement runs for roughly  $100/13 \approx 7.7$  times as many iterations as the  $O(\log n)$  asymptotic model. Accordingly, the tightest correct statement of the runtime theorem uses  $\mathbf{niter}$  explicitly as a constant multiplier.

**Theorem 2** (Runtime bound). *METIS recursive bisection for the balanced  $k$ -cut redistricting problem on a planar graph with  $n$  vertices runs in  $O(\mathbf{niter} \cdot n \log k)$  time with  $O(n \log k)$  space, where  $\mathbf{niter}$  is the number of FM refinement passes. With the experimental setting  $\mathbf{niter} = 100$ , this gives  $O(n \log k)$  up to a constant of 100.*

### 4.2 Empirical scaling across 50 states

Table 1 reports measured runtimes for representative states from the 2020 Census. The range for multi-district states spans from Iowa (1,542 tracts, 4 districts) to California (8,057 tracts, 52 districts), covering a  $5\times$  range in  $n$  among the states shown.<sup>1</sup>

Fitting  $\log T = \log a + b \log n$  by OLS on the  $k > 1$  subset (43 states, 2020 Census) gives  $b = 1.07 \pm 0.03$  ( $R^2 = 0.984$ ), confirming near-linear scaling. The residual super-linearity (0.07 above linear) is attributable to the  $\log k$  factor: larger states tend to have more districts, so both  $n$  and  $k$  grow together.

---

<sup>1</sup>Single-district states ( $k = 1$ ) are excluded from the scaling analysis. The seven excluded states are: Alaska, Delaware, Montana, North Dakota, South Dakota, Vermont, and Wyoming. For  $k = 1$  there is no bisection step; the algorithm performs only adjacency graph construction and returns immediately. Runtime for these states is approximately 2 ms regardless of  $n$ , consistent with  $O(n)$  graph-building time and a negligible constant. Including them in the  $T \sim n^b$  regression would bias the exponent estimate by anchoring low- $n$  observations at near-zero runtimes. The remaining 43 states ( $k \geq 2$ ) form the OLS sample.

Table 1: Empirical runtime scaling on 2020 Census tract data (mean of 10 runs  $\pm$  std. dev., single-threaded, AMD Ryzen 9 5950X). Single-district states ( $k = 1$ ) excluded; see text.

| State      | Tracts ( $n$ ) | Districts ( $k$ ) | Runtime (ms)    | ms/tract |
|------------|----------------|-------------------|-----------------|----------|
| Iowa       | 1,542          | 4                 | $18.4 \pm 0.6$  | 0.012    |
| Oregon     | 2,049          | 6                 | $28.7 \pm 0.9$  | 0.014    |
| Wisconsin  | 1,921          | 8                 | $31.2 \pm 1.1$  | 0.016    |
| Virginia   | 2,748          | 11                | $47.8 \pm 1.5$  | 0.017    |
| Michigan   | 2,813          | 13                | $51.4 \pm 1.7$  | 0.018    |
| Georgia    | 3,053          | 14                | $58.6 \pm 2.0$  | 0.019    |
| Ohio       | 2,884          | 15                | $53.2 \pm 1.8$  | 0.018    |
| Florida    | 4,245          | 28                | $98.7 \pm 3.2$  | 0.023    |
| New York   | 4,919          | 26                | $121.3 \pm 4.1$ | 0.025    |
| Texas      | 5,265          | 38                | $147.1 \pm 4.8$ | 0.028    |
| California | 8,057          | 52                | $198.4 \pm 5.9$ | 0.025    |

**Log  $k$  covariate test.** To test whether the  $\log k$  term in the theoretical bound  $T = O(n \log k)$  accounts for the residual super-linearity, we extend the regression to include  $\log(\log k)$  as a second covariate:

$$\log T = \log a + b \log n + c \log(\log k).$$

Fitting this two-variable model by OLS on the same 43-state, 2020-Census sample gives  $b = 1.05 \pm 0.03$ ,  $c = 0.08 \pm 0.15$ ,  $R^2 = 0.985$ . The coefficient on  $\log(\log k)$  is not significantly different from zero (95% CI:  $[-0.07, 0.23]$ ;  $p = 0.41$ ). Adding  $\log(\log k)$  as a covariate does not significantly improve fit ( $\Delta\text{-AIC} < 2$ ,  $c = 0.08 \pm 0.15$ ), confirming that  $n$  is the dominant scaling factor. The seven  $k = 1$  states excluded from this analysis are Alaska, Montana, North Dakota, South Dakota, Vermont, Delaware, and Wyoming, as noted in the footnote above. The  $\log k$  range among the 43 included states is narrow ( $\log k \in [0.69, 3.95]$ , i.e.  $k$  from 2 to 52), which limits the power to detect a  $\log k$  effect if it exists — but the point estimate  $c \approx 0.08$  is small relative to  $b \approx 1.05$ , and the CI comfortably includes zero. We conclude that  $n$  alone is the sufficient predictor of runtime at census-tract scale.

### 4.3 Cross-census scaling

Table 2 compares scaling exponents across census years, fitted on  $k > 1$  states only. The exponent is stable across years despite changes in tract count and district assignments.

The consistency across census years ( $b \approx 1.07$  in all three) gives confidence that the algorithm will scale similarly for the 2030 Census, which is expected to enumerate approximately 78,000–82,000 tracts.

### 4.4 Space complexity

The bisection tree requires  $O(n \log k)$  space to store the partition hierarchy, as each of the  $n$  tracts appears in one subproblem at each of the  $\lceil \log_2 k \rceil$  levels.

Table 2: Empirical scaling exponent  $b$  in  $T = a \cdot n^b$  by census year ( $k > 1$  states only; single-district states excluded)

| Census year | $n$ (total tracts, $k > 1$ states) | Scaling exponent $b$ | $R^2$ |
|-------------|------------------------------------|----------------------|-------|
| 2000        | 65,443                             | $1.08 \pm 0.04$      | 0.981 |
| 2010        | 73,056                             | $1.06 \pm 0.03$      | 0.983 |
| 2020        | 74,134                             | $1.07 \pm 0.03$      | 0.984 |

Peak memory during coarsening is  $O(n)$  per subproblem (the coarsened graph never exceeds the original size). Since subproblems at the same level are processed sequentially, total peak memory is:

$$M_{\text{peak}} = O(n \log k).$$

**Numerical example.** For California ( $n = 8,057$ ,  $k = 52$ ),  $\log_2 52 \approx 5.7$ , giving a peak memory requirement of approximately  $8,057 \times 6 \approx 48,000$  node records. At 64 bytes per node record (population, adjacency pointers, partition label), this is under 3 MB — trivial on modern hardware.

For block-group resolution ( $n \approx 239,000$  nationally), peak memory per state is under 20 MB, well within the 16 GB available on the reference workstation.

## 5 Discussion

### 5.1 Implications of NP-hardness for the DIA

Theorem 1 establishes that no polynomial-time algorithm can guarantee an optimal redistricting map unless  $P = NP$ .  $P = NP$  is universally believed to be false among theoretical computer scientists; making the NP-hardness result effectively unconditional for all practical purposes. This has two important implications for the Districting Integrity Act.

First, the DIA’s use of a heuristic (METIS) is not a design weakness but a theoretical necessity: any algorithm that claimed to find the optimal compact map would be implicitly claiming to solve an NP-hard problem. Second, legal challenges that argue the DIA should use the “most compact” possible algorithm are misdirected: finding the globally most compact map is computationally intractable.

It is important to note that Theorem 1 concerns the *optimisation* problem (finding the minimum-cut partition), not feasibility (finding any valid partition). A valid partition always exists for any connected graph with  $k \leq n$ : any spanning tree can be cut into  $k$  connected subtrees. NP-hardness means that selecting the *best* partition is computationally intractable — not that redistricting is impossible.



## 5.2 Approximation quality in practice

Proposition 1 establishes that no known polynomial-time algorithm achieves better than  $O(\log n)$  approximation for balanced  $k$ -partition in general; the best known SDP-based algorithm (ARV [Arora et al., 2009]) achieves  $O(\sqrt{\log n})$  for Sparsest Cut. No formal approximation ratio is claimed for METIS’s heuristic.

Empirically, B.7 [Della-Libera, 2026] shows that the DIA seed is within 3% of the best partition found across 10,000 independent seeds. This empirical gap is not the same as the theoretical approximation ratio relative to the true optimum (which is unknown); the reference point is the best of 10,000 seeds, not a provably optimal solution. The empirical 3% gap should be interpreted as evidence of practical solution quality, not as a formal bound. Legal challenges should note that the worst-case guarantee from complexity theory is much weaker than the empirical performance: METIS systematically finds solutions close to the best feasible result found by exhaustive seed search at the redistricting scale.

## 5.3 Comparison to alternative algorithms

Markov chain approaches [Fifield et al., 2020] can sample from the solution space and find near-optimal maps, but their convergence time is exponential in the worst case. IP-based approaches [Mehrotra et al., 1998] can find exact solutions for small instances ( $n \leq 500$ ) but do not scale to census-tract graphs.

The METIS runtime of  $O(n^{1.07})$  is the best available among algorithms that produce legally compliant (contiguous, population-balanced) maps at census-tract scale. The next fastest alternatives (simulated annealing, genetic algorithms) are  $O(n^{1.5})$  or worse on the same problem class.

## 5.4 Limitations

The NP-hardness reduction in Theorem 1 holds for the optimisation version (find a partition with minimum edge cut subject to connectivity and population balance). The decision version (does a partition with edge cut  $\leq C$  exist?) is co-NP-hard by standard argument. No formal approximation ratio for METIS recursive bisection is established here; tighter bounds specific to the redistricting graph class (sparse planar graphs with bounded diameter and integer population weights) remain an open problem.

The empirical scaling ( $b = 1.07$ ,  $k > 1$  states) is fit on the 50-state sample and may not generalise to block-level resolution ( $n \approx 8.1\text{M}$  nationally), which has not been tested at full scale.

# 6 Conclusion

We established the theoretical and empirical complexity profile of METIS recursive bisection for congressional redistricting.

The key results are:

1. **NP-hardness:** Balanced  $k$ -cut redistricting on planar graphs is NP-hard (Theorem 1), by reduction from Connected Planar Graph Bisection [Dyer and Frieze, 1985]. This correctly accounts for the contiguity requirement of the redistricting problem. The hardness is for optimisation, not feasibility: a valid partition always exists for connected graphs.
2. **Approximation landscape:** No formal approximation ratio is known for METIS’s heuristic (Proposition 1). The best known polynomial-time result ( $O(\sqrt{\log n})$  via ARV [Arora et al., 2009]) applies to Sparsest Cut, not to balanced connected  $k$ -partition. Empirically, METIS is within 3% of the best of 10,000 seeds across redistricting-scale instances [Della-Libera, 2026].
3. **Runtime:**  $O(\text{niter} \cdot n \log k)$  with `niter` = 100 fixed; empirically  $O(n^{1.07 \pm 0.03})$  on  $k > 1$  states across three census years, confirming near-linear practical scaling. Single-district states are excluded from the scaling fit.
4. **Space:**  $O(n \log k)$ , under 3 MB for the largest state (California,  $n = 8,057$ ,  $k = 52$ ) at tract resolution.

These results provide the complexity-theoretic foundation for the DIA’s algorithm specification. NP-hardness establishes that exact optimisation is intractable, making heuristic approximation not merely acceptable but necessary. The empirical 3% gap from the best feasible seed provides a practical bound on solution quality for legal and policy purposes.

**Future work.** Open problems include: (1) a tighter approximation ratio specific to the redistricting graph class, (2) empirical scaling at block-level resolution ( $n \approx 8.1\text{M}$ ), and (3) complexity analysis of the multi-constraint variant (VRA compliance + county preservation).

## References

- Konstantin Andreev and Harald Räcke. Balanced graph partitioning. *Theory of Computing Systems*, 39(6):929–939, 2006.
- Sanjeev Arora, Satish Rao, and Umesh Vazirani. Expander flows, geometric embeddings and graph partitioning. *Journal of the ACM*, 56(2):5:1–5:37, 2009.
- Gio Della-Libera. Solution space and seed sensitivity in METIS redistricting, 2026. B.7 in Algorithmic Redistricting series.
- Moon Duchin and Bridget Eileen Tenner. Discrete geometry for electoral geography. *arXiv*, 2020. arXiv:1808.05860.
- Martin E. Dyer and Alan M. Frieze. On the complexity of partitioning graphs into connected subgraphs. *Discrete Applied Mathematics*, 10(2):139–153, 1985. doi: 10.1016/0166-218X(85)90008-3.

- Charles M. Fiduccia and Robert M. Mattheyses. A linear-time heuristic for improving network partitions. *Proc. 19th Design Automation Conference (DAC)*, pages 175–181, 1982.
- Benjamin Fifield, Kosuke Imai, Jun Kawahara, and Christopher T. Kenny. The essential role of empirical validation in legislative redistricting simulation. *Statistics and Public Policy*, 7(1):52–68, 2020.
- Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W.H. Freeman, 1979.
- Michael R. Garey, David S. Johnson, and Larry Stockmeyer. Some simplified NP-complete graph problems. *Theoretical Computer Science*, 1(3):237–267, 1976. doi: 10.1016/0304-3975(76)90059-1.
- Olivier Goldschmidt and Dorit S. Hochbaum. A polynomial algorithm for the  $k$ -cut problem for fixed  $k$ . *Mathematics of Operations Research*, 19(1):24–37, 1994.
- George Karypis and Vipin Kumar. A fast and high quality multilevel scheme for partitioning irregular graphs. *SIAM Journal on Scientific Computing*, 20(1):359–392, 1998a. doi: 10.1137/S1064827595287997.
- George Karypis and Vipin Kumar. Multilevel  $k$ -way partitioning scheme for irregular graphs. volume 48, pages 96–129, 1998b.
- Richard J. Lipton and Robert E. Tarjan. A separator theorem for planar graphs. *SIAM Journal on Applied Mathematics*, 36(2):177–189, 1979.
- Anuj Mehrotra, Ellis L. Johnson, and George L. Nemhauser. An optimization based heuristic for political redistricting. *Management Science*, 44(8):1100–1114, 1998.
- Sebastian Schlag, Tobias Heuer, Lars Gottesbüren, Yaroslav Akhremtsev, Christian Schulz, and Peter Sanders.  $k$ -way hypergraph partitioning via  $n$ -level recursive bisection. In *Proc. 19th SIAM Workshop on Algorithm Engineering and Experiments (ALENEX)*, pages 53–67, 2017.